Phase 3: Generic Fantasy RPG Mechanics Design

⊠ Current State Summary

We've successfully deployed a **working AI GM backend** with:

- Campaign/character/turn persistence in SQLite
- `turn_number` per-campaign sequencing (1,2,3,4)
- Gemma3:4b generating coherent Polish fantasy narrative
- `/api/campaigns/{id}/turns` flow operational

Campaign 16 proof-of-concept demonstrates full conversation:

```
Turn 1: Klasztor Krwawego Kamienia (setup)
Turn 2-3: Room description (grzyby, portal, pergamin)
Turn 4: Pergamin pickup → dzidy encounter ("Co robisz?")
```

⊠ Phase 3 Scope: Generic Fantasy Mechanics Template

Design **system-agnostic fantasy RPG mechanics** that work across Warhammer Fantasy, D&D, Pathfinder, etc., stored in `character.sheet_json` and fed into LLM context.

Core Mechanics to Define

```
1. **CHARACTER CREATION**
   ├── Base stats (STR, DEX, CON, INT, WIS, CHA)
   ├── Derived stats (HP, AC, Initiative, Speed)
   ├── Skills/Proficiencies
   ├── Equipment/Inventory
   ├── Level/Experience
   └── Saving throws

2. **COMBAT SYSTEM**
   ├── Initiative order
   ├── Attack rolls (d20 + modifiers)
   ├── Damage calculation
   ├── Armor Class / Defense
   ├── Conditions (poisoned, prone, etc.)
   └── Death saves

3. **PROGRESSION**
   ├── XP thresholds per level
   ├── Stat increases
```

```
   ├── New abilities/spells
   ├── Proficiency bonuses

4. **RPG SYSTEMS INTEGRATION**
   ├── `/roll 1d20+5` → parsed dice + narrated results
   ├── `/sheet` → current character state
   ├── `/status` → party/campaign overview
   └── `/heal 1d8` → auto HP updates

5. **WORLD STATE TRACKING**
   ├── Locations (current room, explored areas)
   ├── NPCs (hostile/friendly tracking)
   ├── Inventory (weight, slots)
   └── Quest log
```

Discussion Points for Generic Fantasy Template

1. Character Sheet Structure

```
{
  "level": 1,
  "experience": 0,
  "stats": {
    "str": 15, "dex": 14, "con": 13, "int": 12, "wis": 10, "cha": 8
  },
  "hp": {"max": 12, "current": 12},
  "ac": 14,
  "initiative": "+2",
  "speed": 30,
  "skills": {"stealth": "+4", "perception": "+2"},
  "inventory": ["miecz", "skórzany pas", "srebrny medalion"],
  "status": [],
  "location": "Klasztor Krwawego Kamienia"
}
```

Questions:

- Flat stats vs ability score modifiers (STR 15 = +2)?
- HP tracking per character? Party sheet?
- Skill list: fixed generic (Athletics, Arcana) or system-specific?

2. Combat Mechanics

```
Attack: d20 + STR mod + proficiency ≥ AC → hit
Damage: weapon dice + STR mod
```

Questions:

- Generic weapon table (miecz=1d8, łuk=1d6+DEX)?

- Critical hits (20 auto-hit, double damage)?
- Advantage/disadvantage (2d20 take highest/lowest)?

3. Dice Commands

Current `/commands/execute` exists but unused. Extend `/turns`:

```
User: "Atakuję dzidę /roll 1d20+3"
AI: "Rzut: 17+3=20 > AC14 → trafienie! /roll 1d8+2 → 6 obrażeń"
```

Questions:

- Parse `/roll NdM+X` inline in `user_text`?
- Store roll results in `turn.assistant_text`?
- Critical success/failure narration?

4. Progression Template

```
Level 1: 0 XP
Level 2: 300 XP
Level 3: 900 XP
+1 stat every 4 levels
```

Questions:

- XP from combat (CR-based) + quests?
- Auto-levelup or manual?
- Generic spell progression?

5. World State Management

Party sheet in `campaign.sheet_json`:

```
{
  "active_characters": [16, 17],
  "initiative_order": [],
  "current_encounter": {
    "enemies": [{"name": "dzida", "hp": 8, "ac": 12}],
    "round": 1
  },
  "quests": ["Znajdź pochodzenie dzid"]
}
```

Proposed Next Steps

Week 1: Character Sheet v1

```
1. Define generic fantasy stats in CharacterCreateRequest
2. `/sheet` → full JSON dump
3. `/status` → party summary
4. Store in `character.sheet_json`
```

Week 2: Dice Parser

```
1. Extend `/turns` to parse `/roll NdM+X`
2. Return results in `assistant_text`
3. Feed rolls into LLM context
```

Week 3: Combat State

```
1. Party sheet in campaign
2. Initiative tracking
3. HP updates from dice rolls
```

Your Priorities?

1. **Character creation first** (sheet structure)?
2. **Dice integration** (`/roll` commands)?
3. **Combat state** (initiative, HP tracking)?
4. **Frontend** (show sheets, rolls in UI)?

What mechanics matter most for your gaming style? Generic fantasy template gives us foundation for Warhammer/Cyberpunk later.

Ready to spec character sheet JSON?